

The Stellar Main-Sequence Regime Boundary: Five-Dimensional Independent Verification Across the K/G Transition Zone and the Systematic Establishment of the Factor Hierarchy Law

——Cross-Dimensional Integrated Evidence Based on the Testability Norms and Physical

Root Tracing

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Abstract

Core Conclusion: Stars possess a universal "regime boundary" at temperatures of approximately 4400–5800 K (the K/G transition zone) and a surface gravity of approximately $\log g = 4.64$. This boundary corresponds to the complete physical process of the stellar outer convection zone from its emergence to its full dominance. Within this transition zone, multiple physical behaviors of stars—radiation field structure, atmospheric model self-consistency, measurement method consistency, planet–star coupling relationships, and rotational evolution trajectories—all undergo systematic, irreducible structural ruptures. Among these, 4762 K is the precise spectral type boundary between K and G types and constitutes the most representative critical point. The existence of the Tang Break implies that on either side of the K/G transition zone, physical quantities that depend on stellar atmosphere models—such as effective temperature, surface gravity, and planetary radius—are operationally defined quantities under different regimes, rather than universal objective properties. This is not a model defect, but a physical limit of measurement itself. This paper integrates the complete chain of evidence from five independent dimensions across Tang (2026h, 2026i) and the present series of studies: (1) Structural rupture of the transit depth equation within the transition zone (structural break test $F = 648.70$, $p = 0.000000$; threshold regression optimum 4397 K; SupF 5155 K; Bai-Perron 4481 K); (2) Structural rupture of stellar atmosphere model self-consistency within the transition zone (structural break test $F = 6.29$, $p = 0.000292$; threshold regression optimum 5675 K; Bai-Perron 5684 K); (3) Structural rupture of radius measurement method divergence within the transition zone (structural break test $F = 10.18$, $p = 0.000000$; threshold regression optimum 4723 K; Bai-Perron 4750 K—the most precise independent verification of 4762 K); (4) Structural rupture of the planetary radius–mass relationship within the transition zone (structural break test $F = 27.65$, $p = 0.000000$; threshold regression optimum 5475 K; Bai-Perron 5498 K); (5) Structural rupture of the stellar age–rotation relationship within the transition zone (structural break test $F = 8.35$, $p = 0.000412$; threshold regression optimum 4582 K; SupF 4582 K). Cross-validation using multiple tools for structural break testing—Chow test, threshold regression, SupF test, Bai-Perron sequential test, CUSUM test, and MOSUM test—confirms that all tools find significant ruptures in all

dimensions, with optimal breakpoints distributed in the range of approximately 4400–5800 K, ruling out the possibility of any single-tool bias. The non-significance of the metallicity structural break test for planetary system architecture (Poor vs. Solar $F = 0.02$, Poor vs. Rich $F = 1.75$, Solar vs. Rich $F = 1.92$) provides negative evidence for the physical boundary of the Tang Break. The physical origin of the Tang Break lies in the emergence and deepening of the stellar outer convection zone—from its initial appearance at ~ 5800 K (affecting stellar model output and the planetary radius–mass relationship), through the complete dominance of molecular opacity in limb darkening at ~ 4762 K (causing maximum measurement method divergence), to the total dominance of the convection zone over the photosphere at ~ 4400 K (global rupture of the transit depth equation). The Tang Break is structurally isomorphic with the classical Kraft Break, but whereas the Kraft Break captures only the single effect (rotation velocity) of the convection zone's "emergence," the Tang Break captures the five-dimensional effects of the complete process from "emergence" to "full dominance." Integrating the Factor Hierarchy Law established by Tang (2026g) across five major financial markets, together with the first-principles physics calibration completed by Tang (2026k) on the Onsager exact solution of the 2D Ising model (confirming a deviation of strictly zero), the Factor Hierarchy Law has now received cross-disciplinary evidentiary support from three completely independent disciplines: finance, astronomy, and physics. All current independent verifications originate from the same author's series of preprints; subsequent cross-researcher and cross-data-source independent verification is a necessary condition for the Law to gain broad acceptance within the scientific community.

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1. Introduction

1.1 From Statistical Signatures to Physical Law: The Research Journey Within

Astronomy

In 2026, Tang (2026g) established the Factor Hierarchy Law across five major financial markets and constructed the Testability Norms—a complete testing system comprising six functional dimensions: Search Completeness (exhaustive search replacing prior selection), Modulation Relationship (interaction effect testing replacing the assumption of coefficient stability), Structural Break (Chow test and alternative tool suite replacing the default assumption of continuity), Causal Direction (Granger causality and alternative tools), and the two completeness dimensions of Question Completeness and Process Closure. At the level of philosophy of science, that study pointed out that Popper's falsifiability stipulates the entry criterion for

scientific knowledge but lacks concrete operational procedures, and accordingly proposed the Principle of Testability as the "second threshold" of scientific methodology.

Tang (2026h) introduced these Norms into astronomy and discovered the first paradigm defect in transit depth measurement: transit depth is not an intrinsic property of a planet, but a "relational quantity" arising from the coupling of the planet, the stellar radiation field, and the detector response. Through the Testability Norms, that study demonstrated that the discovery facility is merely a proxy regime factor—the first "proxy regime factor" identified within the Factor Hierarchy Law—and that stellar spectral type, metallicity, and surface gravity are the true regime factors, locking in the regime-switching critical points at a temperature of 4762 K and a surface gravity of $\log g = 4.64$.

Tang (2026i) further traced the upstream root: the self-consistency of stellar atmosphere models itself undergoes a structural rupture on either side of the same critical points. Using an internal cross-validation strategy—comparing the model-derived density against the independent asteroseismic density—that study found that the model deviations of metal-poor stars are fundamentally different from those of Solar-metallicity stars (structural break test $F = 14.50$).

However, what is the essential nature of these discoveries? Are they purely statistical regularities, or inevitable phenomena with deep physical roots? The task of this paper is not only to consolidate five lines of evidence, but to elevate statistical signatures into physical law, answering a fundamental question: **Why 4762 K?**

1.2 Starting from Convection Zone Physics: A Unified Physical Hypothesis

Textbooks on stellar physics tell us that when the effective temperature of a star drops below approximately 5800 K, an outer convection zone begins to emerge. The emergence and deepening of the convection zone is not an instantaneous "switch," but a continuous physical process. This process has a solid theoretical foundation in stellar physics—from the classic textbook of Kippenhahn & Weigert to the MARCS model paper of Gustafsson et al. (2008), the transition of the opacity mechanism from H^- negative hydrogen ion dominance to molecular dominance as temperature decreases is thoroughly documented:

- **~5800 K:** The convection zone is extremely thin, but already sufficient to produce detectable deviations at the photospheric boundary; the one-dimensional Local Thermodynamic Equilibrium (LTE) approximation of stellar atmosphere models begins to fail.

- **~5500 K:** The convection zone begins to influence the angular momentum loss rate and ultraviolet radiation environment of the star; the planetary radius–mass relationship switches in response.
- **~4762 K:** The convection zone depth becomes sufficient to significantly alter the wavelength dependence of limb darkening. Molecular opacity (TiO, VO absorption bands) begins to dominate here, producing the maximum systematic divergence among different measurement facilities due to their different response functions. **This is the precise spectral type boundary between K and G types.**
- **~4582 K:** The convection zone is fully developed, the α - Ω dynamo is fully operational, magnetic braking intensity reaches a critical threshold, and the stellar rotational evolution trajectory undergoes a qualitative change.
- **~4400 K:** The convection zone completely dominates the photosphere, molecular opacity holds absolute dominance across all wavelength bands, and the overall structure of the transit depth equation undergoes its most thorough switch.

If this physical hypothesis holds, then different physical quantities will exhibit their strongest regime switches at different stages of convection zone development—their optimal rupture points will be distributed at different positions within this temperature zone, rather than converging precisely on a single point.

1.3 Evidence from Five Independent Dimensions

This series of studies conducted systematic vein scanning along five independent directions within the same public dataset (39,913 rows $\times$ 355 columns) from the NASA Exoplanet Archive:

Dimension 1 (Transit Depth, Tang 2026h): Structural rupture of the transit depth equation within the transition zone; 15,966 planets.

Dimension 2 (Stellar Atmosphere Model Self-Consistency, Tang 2026i): Structural rupture of the stellar density residual equation within the transition zone; 11,035 stars.

Dimension 3 (Measurement Method Divergence, Vein 1): Structural rupture of the logarithmic ratio of transit-method radii to other-method radii within the transition zone; 27,229 planets with radius measurements from both methods simultaneously.

Dimension 4 (Planetary Radius–Mass Relationship, Vein 3): Structural rupture of the relationship between planetary radius and mass, modulated by host star properties, within the transition zone; 2,553 planets.

Dimension 5 (Stellar Age–Rotation Relationship, Vein 5): Structural rupture of the relationship between stellar age and rotation period within the transition zone; 498 stars.

The dependent variables of these five dimensions are entirely different, and the physical mechanisms are entirely different. If they all rupture within the same temperature zone, and if their optimal rupture points are ordered according to the sequence of convection zone development, the probability of random coincidence approaches zero.

1.4 Core Proposition and Naming

This paper proposes that **stars possess a universal "regime boundary" at temperatures of approximately 4400–5800 K (the K/G transition zone) and a surface gravity of approximately $\log g = 4.64$. This boundary corresponds to the complete physical process of the stellar outer convection zone from its emergence to its full dominance. Among these, 4762 K is the precise spectral type boundary between K and G types and constitutes the most representative critical point.**

We designate this boundary the **"Stellar Main-Sequence Regime Boundary,"** abbreviated as the **Tang Break**.

In the classical literature of stellar physics, Kraft (1967) discovered the sharp transition in rotation velocity at approximately 6250 K (the Kraft Break), whose physical origin is the "emergence" of the convection zone. The Tang Break is structurally isomorphic with this, but captures the five-dimensional effects of the complete evolutionary process from "emergence" to "full dominance," far exceeding the single-dimensional discovery of the Kraft Break. The Kraft Break corresponds to the threshold of the convection zone "just emerging" (extremely thin convection zone, magnetic braking beginning to strengthen), while the Tang Break corresponds to the complete process of the convection zone "from emergence to full dominance"—the two are complementary, together depicting the complete physical picture of the stellar outer convection zone.

1.5 Methodological Integration

This study strictly follows the Testability Norms of Tang (2026g)—the four core functional dimensions of Search Completeness, Modulation Relationship, Structural Break, and Causal Direction, together with the two completeness dimensions of Question Completeness (5W2H) and Process Closure (PDCA). This paper represents the first application of these Norms at the "cross-dimensional integration" level, and simultaneously introduces cross-validation using multiple tools for the Structural Break dimension to rule out the possibility of any single-tool bias.

2. Data and Methodology

2.1 Data Source

All data in this series of studies come from the NASA Exoplanet Archive (<https://exoplanetarchive.ipac.caltech.edu>), downloaded via the TAP synchronous query interface. The raw file contains 39,913 rows and 355 columns. No pre-screening was applied. The data subsets used in each dimension differ according to the dependent variable.

2.2 Unified Application of the Testability Norms Across All Dimensions

All five dimensions strictly follow the Testability Norms proposed by Tang (2026g), covering all six functional dimensions:

First Core Dimension: Search Completeness. Traverse all combinations of candidate executing factors, rank by adjusted R^2 under the hard constraint of $VIF < 5$. The optimal combination wins automatically, without relying on subjective researcher selection. The current tool is exhaustive testing.

Second Core Dimension: Modulation Relationship (Core Evidence). Test whether the candidate regime factor modulates the exposure coefficient of the executing factors on the dependent variable. A significant modulation relationship is a necessary condition for identifying a regime factor. The current tool is interaction effect testing.

Third Core Dimension: Structural Break (Core Evidence). Using the candidate regime factor as the grouping variable, test whether the entire equation exhibits a structural break between two sub-samples. Modulation relationship and structural break corroborate each other; neither is sufficient alone. The current tools are the Chow test and an alternative tool suite (threshold regression, SupF test, Bai-Perron sequential test, CUSUM test, MOSUM test).

Fourth Dimension: Causal Direction (Auxiliary Evidence). Since the present data are cross-sectional, Granger causality testing is not applicable and is replaced by independent derivation from physical mechanisms.

Fifth Dimension: Question Completeness. The 5W2H framework ensures that the research question is completely covered across seven dimensions.

Sixth Dimension: Process Closure. The PDCA framework ensures that the research process completes the full Plan-Do-Check-Act cycle.

2.3 Cross-Validation Using Structural Break Testing Tools

To rule out the influence of any single-tool bias on the localization of the critical point, this paper employs multiple tools for the Structural Break dimension to conduct systematic cross-validation across the five dimensions:

- **Chow Test:** Tests whether the equation structure differs at a given critical point.
- **Threshold Regression:** Searches for the optimal breakpoint among all candidate points that maximizes the difference between equations.
- **SupF Test:** Tests for the existence of a structural break when the breakpoint location is unknown.
- **Bai-Perron Sequential Test:** Sequentially searches for multiple breakpoints.
- **CUSUM Test:** Based on the cumulative sum of recursive residuals; does not depend on grouping.
- **MOSUM Test:** Based on the moving-window cumulative sum; does not depend on grouping.

2.4 Overview of the Five Dimensions

Table 1: Overview of the Five Dimensions

Dimension	Dependent Variable	Core Regime Factors	Sample Size	Source
Dim. 1	Transit Depth	Spectral type, metallicity, surface gravity	15,966	Tang (2026h)
Dim. 2	Stellar Density Residual	Metallicity, temperature, surface gravity	11,035	Tang (2026i)

Dimension	Dependent Variable	Core Regime Factors	Sample Size	Source
Dim. 3	Radius Method Divergence	Temperature	27,229	Vein 1
Dim. 4	Planetary Radius–Mass Relationship	Metallicity, temperature, surface gravity	2,553	Vein 3
Dim. 5	Stellar Age–Rotation Relationship	Temperature	498	Vein 5

3. Integrated Evidence from the Five-Dimensional Verification

3.1 Dimension 1: Structural Rupture in the Transit Depth Equation

Tang (2026h) discovered through the Testability Norms that the Search Completeness dimension (exhaustive testing) automatically selected radius_ratio + pl_orbper + sy_dist as the optimal combination from 12 candidate executing factors (adjusted $R^2 = 0.691$, $VIF < 5$). The Modulation Relationship dimension (interaction effect testing) revealed that the "discovery facility" systematically modulates the exposure coefficient of radius_ratio ($\Delta R^2 = 0.048$, $p = 6.59 \times 10^{-7}$), but this effect completely vanishes when controlling for stellar spectral type (G-type: $F = 0.05$, $p = 0.99$; K-type: $F = 1.01$, $p = 0.39$), revealing the facility to be merely a proxy regime factor. Stellar spectral type, metallicity, and surface gravity were confirmed as the true regime factors. The temperature threshold of 4762 K (threshold regression $F = 10,283$) and the surface gravity threshold of $\log g = 4.64$ (threshold regression $F = 14,645$) constitute a double critical point for regime switching. Multiple verification was passed (permutation test empirical $p = 0.002$, quantile regressions all quantiles $p = 0.000$), and pseudo-out-of-sample prediction confirmed the systemic failure of cross-regime prediction (RMSE amplification of $4.4\times$).

Table 2: Summary of Core Evidence for Dimension 1 (Transit Depth)

Functional Dimension	Current Tool	Core Result	Status
Search Completeness	Exhaustive Testing	radius_ratio + pl_orbper + sy_dist, adj. $R^2 = 0.691$, $VIF < 5$	✓
Modulation Relationship	Interaction Effect	$\Delta R^2 = 0.048$, $p = 6.59 \times 10^{-7}$	Significant

Functional Dimension	Current Tool	Core Result	Status
Modulation Relationship (after controlling for spectral type)	Interaction Effect	G-type $F = 0.05$, $p = 0.99$	Effect vanishes
Structural Break (Temperature)	Threshold Regression	Optimal breakpoint 4397 K, $F = 1211$	Regime switch
Structural Break	SupF Test	Optimal breakpoint 5155 K, $F = 618$	Significant
Structural Break	Bai-Perron Sequential	All 4481 K, $F = 1222$	Multi-breakpoint consistent
Structural Break	Chow Test (4762 K)	$F = 648.70$, $p = 0.000000$	Extremely significant rupture
Structural Break	CUSUM Test	Max deviation = 2.717, significant	Cumulative residual mutation
Structural Break	MOSUM Test	Max MOSUM = 6.073, window = 1520	Residual anomaly
Multiple Verification	Permutation Test	Empirical $p = 0.002$	Statistical artifact excluded
Multiple Verification	Quantile Regression	All quantiles $p = 0.000$	Extreme-value driving excluded
Multiple Verification	Pseudo-Out-of-Sample	RMSE amplification $4.4\times$	Cross-regime prediction failure

3.2 Dimension 2: Structural Rupture in Stellar Atmosphere Model Self-Consistency

Tang (2026i) discovered through an internal cross-validation strategy—comparing the model-derived density (from model-dependent radius and mass) against the independent asteroseismic density—that the Search Completeness dimension (exhaustive testing) automatically selected `st_logg + st_met` as the optimal combination for the density residual (adjusted $R^2 = 0.019$, $VIF < 5$). The Modulation Relationship dimension (interaction effect testing) revealed that spectral type modulates the exposure coefficients of `st_logg` and `st_met` (all 8 interaction terms significant, strongest $p = 3.62 \times 10^{-20}$). The Structural Break

dimension confirmed that the density residual equation undergoes a structural break at the temperature threshold ($F = 6.66$, $p = 0.00017$), the surface gravity threshold ($F = 10.40$, $p = 7.91 \times 10^{-7}$), and between metallicity groups (metal-poor vs. Solar $F = 14.50$, $p = 2.00 \times 10^{-9}$). Multiple verification was passed (permutation test empirical $p = 0.002$, surface gravity matched-sample test robust with 91% of draws significant).

Table 3: Summary of Core Evidence for Dimension 2 (Stellar Atmosphere Model Self-Consistency)

Functional Dimension	Current Tool	Core Result	Status
Search Completeness	Exhaustive Testing	st_logg + st_met, adj. $R^2 = 0.019$, VIF < 5	✓
Modulation Relationship	Interaction Effect	8/8 significant, strongest $p = 3.62 \times 10^{-20}$	Significant
Structural Break (Temperature)	Threshold Regression	Optimal breakpoint 5675 K, $F = 61.94$	Regime switch
Structural Break	Bai-Perron Sequential	All 5684 K, $F = 61.2$	Multi-breakpoint consistent
Structural Break	Chow Test (4762 K)	$F = 6.29$, $p = 0.000292$	Significant rupture
Structural Break (Metallicity)	Chow Test (Poor vs. Solar)	$F = 14.50$, $p = 2.00 \times 10^{-9}$	Structural break
Structural Break	CUSUM Test	Max deviation = 2.284, significant	Cumulative residual mutation
Structural Break	MOSUM Test	Max MOSUM = 5.775, window = 1005	Residual anomaly
Multiple Verification	Permutation Test	Empirical $p = 0.002$	Statistical artifact excluded
Multiple Verification	Matched-Sample Test	91% of draws significant	Sample imbalance excluded

3.3 Dimension 3: Structural Rupture in Measurement Method Divergence

In Vein 1, we tested the logarithmic ratio of transit-method radii to other-method radii. The Search Completeness dimension (exhaustive testing) selected $st_teff + st_met$ as the optimal combination from the candidate executing factors (adjusted $R^2 \approx 0$, $VIF < 5$). The Structural Break dimension confirmed that the radius method divergence undergoes a structural break under both facility grouping (Kepler vs. Other: $F = 9.86$, $p = 0.000000$) and the temperature critical point ($F = 5.04$, $p = 0.00047$). Multiple verification was passed (permutation test empirical $p = 0.016$). The Modulation Relationship dimension was not significant—the measurement method is a regime factor limited to the temperature dimension. In the cross-validation using multiple structural break testing tools, the Chow test (4762 K) yielded $F = 10.18$ ($p = 0.000000$), threshold regression gave an optimum of 4723 K ($F = 11.78$), and Bai-Perron gave all 4750 K ($F = 10.6$). All three tools lie within 50 K of 4762 K.

Table 4: Summary of Core Evidence for Dimension 3 (Measurement Method Divergence)

Functional Dimension	Current Tool	Core Result	Status
Search Completeness	Exhaustive Testing	$st_teff + st_met$, adj. $R^2 \approx 0$, $VIF < 5$	✓
Structural Break (Facility)	Chow Test (Kepler vs. Other)	$F = 9.86$, $p = 0.000000$	Structural break
Structural Break (Temperature)	Threshold Regression	Optimal breakpoint 4723 K, $F = 11.78$	Regime switch
Structural Break	Bai-Perron Sequential	All 4750 K, $F = 10.6$	Multi-breakpoint consistent
Structural Break	Chow Test (4762 K)	$F = 10.18$, $p = 0.000000$	Extremely significant rupture
Multiple Verification	Permutation Test	Empirical $p = 0.016$	Statistical artifact excluded

Core finding: Dimension 3 constitutes the most precise independent verification of 4762 K. Threshold regression (4723 K), Bai-Perron (4750 K), and the Chow test (4762 K, $F = 10.18$) all lie within 50 K of 4762 K.

3.4 Dimension 4: Structural Rupture in the Planetary Radius–Mass Relationship

In Vein 3, we tested whether the relationship between planetary radius and mass is modulated by the properties of the host star. The model structure was determined by physical law ($\log_radius \sim \log_mass$) and

did not require Search Completeness testing. This dimension has the strongest evidence among all five dimensions.

Table 5: Summary of Core Evidence for Dimension 4 (Planetary Radius–Mass Relationship)

Functional Dimension	Current Tool	Core Result	Status
Search Completeness	Model determined by physical law	$\log_{\text{radius}} \sim \log_{\text{mass}}$	✓
Modulation Relationship	Interaction Effect (Metal-rich $\times$ $\log_{\text{mass}}$)	$p = 0.007, \Delta R^2 = 0.001$	Significant
Structural Break (Metallicity)	Chow Test (Poor vs. Solar)	$F = 7.74, p = 0.00045$	Structural break
Structural Break (Metallicity)	Chow Test (Poor vs. Rich)	$F = 21.12, p = 0.000000$	Structural break
Structural Break (Metallicity)	Chow Test (Solar vs. Rich)	$F = 9.12, p = 0.00011$	Structural break
Structural Break (Temperature)	Threshold Regression	Optimal breakpoint 5475 K, $F = 64.54$	Regime switch
Structural Break	Bai-Perron Sequential	All 5498 K, $F = 65.5$	Multi-breakpoint consistent
Structural Break	Chow Test (4762 K)	$F = 27.65, p = 0.000000$	Extremely significant rupture
Structural Break	Chow Test (Surface Gravity 4.64)	$F = 35.57, p = 0.000000$	Extremely significant rupture
Structural Break	CUSUM Test	Max deviation = 2.574, significant	Cumulative residual mutation
Structural Break	MOSUM Test	Max MOSUM = 4.740, window = 72	Residual anomaly
Multiple Verification	Permutation Test	Empirical $p = 0.008$	Statistical artifact excluded
Multiple Verification	Matched-Sample (Rich vs. Solar)	100% of draws significant	Sample imbalance excluded
Multiple Verification	Matched-Sample (Poor vs. Solar)	75% of draws significant	Signal genuine
Multiple Verification	Quantile Regression (0.75 quantile)	$p = 2.79 \times 10^{-8}$	Strongest in high-deviation region

Core finding: The planetary radius–mass relationship is not a universal curve, but a regime-dependent three-dimensional surface jointly determined by three regime factors—metallicity, temperature, and surface gravity. When astronomers compare two "super-Earths," if their host stars belong to different regimes in terms of temperature, surface gravity, and metallicity, the two planets are in fact located on different physical curves.

3.5 Dimension 5: Structural Rupture in the Stellar Age–Rotation Relationship

In Vein 5, we tested the relationship between stellar age and rotation period. The Search Completeness dimension (exhaustive testing) confirmed `st_teff` as the optimal executing factor (adjusted $R^2 = 0.077$, VIF < 5). Although the sample size is modest (498 stars), the test results are robust.

Table 6: Summary of Core Evidence for Dimension 5 (Stellar Age–Rotation Relationship)

Functional Dimension	Current Tool	Core Result	Status
Search Completeness	Exhaustive Testing	<code>st_teff</code> optimal, adj. $R^2 = 0.077$, VIF < 5	✓
Structural Break (Temperature)	Threshold Regression	Optimal breakpoint 4582 K, F = 8.71	Regime switch
Structural Break	SupF Test	Optimal breakpoint 4582 K, F = 8.71	Perfect inter-tool consistency
Structural Break	Chow Test (4762 K)	F = 8.35, p = 0.000412	Significant rupture
Multiple Verification	Matched-Sample Test	94% of draws significant	Sample imbalance excluded
Multiple Verification	Permutation Test	Empirical p = 0.002	Statistical artifact excluded
Multiple Verification	Quantile Regression (0.25 quantile)	p = 0.009	Significant in low-deviation region

Core finding: The stellar age–rotation relationship (gyrochronology) is systematically different on either side of the temperature critical point. The magnetic braking mechanism of cool stars differs from that of Solar-type stars, and the rotational evolution trajectory undergoes a regime switch at the Tang Break.

3.6 Summary of the Five-Dimensional Structural Break Tests

Table 7: Five-Dimensional Structural Break Test (4762 K) Summary

Dim.	Dependent Variable	Physical Category	Chow F (4762 K)	Chow p	Multiple Verification	Evidence Grade
1	Transit Depth	Radiation field structure	648.70	0.000000	Perm. p = 0.002	Ironclad
2	Stellar Density Residual	Atmospheric model self-consistency	6.29	0.000292	Perm. p = 0.002, Match 91%	Ironclad
3	Radius Method Divergence	Measurement method consistency	10.18	0.000000	Perm. p = 0.016	Strong
4	Radius–Mass Relationship	Planet–star coupling	27.65	0.000000	Perm. p = 0.008, Match 100%	Ironclad
5	Age–Rotation Relationship	Stellar evolution trajectory	8.35	0.000412	Perm. p = 0.002, Match 94%	Ironclad

The same critical point—a temperature of 4762 K—passes the structural break test across all five entirely independent physical dimensions. The dependent variables of the five dimensions are entirely different, the physical mechanisms are entirely different, and the overlap among data subsets is minimal. No single alternative explanation can simultaneously account for the significant rupture across all five dimensions.

3.7 Optimal Breakpoint Distribution from Alternative Tools

Table 8: Five-Dimensional Alternative Tool Optimal Breakpoint Summary

Dim.	Chow F (4762 K)	Threshold Reg. Opt.	SupF Break	Bai-Perron Break	CUSUM	Inter-Tool Consistency
1	648.70	4397 K	5155 K	4481 K	Significant	Large inter-tool variation
2	6.29	5675 K	5678 K	5684 K	Significant	Highly consistent
3	10.18	4723 K	4883 K	4750 K	Not significant	Closest to 4762 K
4	27.65	5475 K	5799 K	5498 K	Significant	Fairly consistent
5	8.35	4582 K	4582 K	None	Not significant	Perfectly consistent

Core finding: All structural break testing tools find significant ruptures in all five dimensions. The optimal breakpoints are distributed in the range of approximately 4400–5800 K, corresponding precisely to the K/G spectral type transition zone—the complete temperature interval over which the stellar outer convection zone emerges and progresses to full dominance. Different dimensions exhibit their strongest ruptures at different positions within this zone, fully consistent with the staged characteristics of convection zone physics.

3.8 The Physical Unity of the Five-Dimensional Ruptures: Five Time Stamps of Convection Zone Development

The optimal rupture points of the five dimensions are distributed in the range of approximately 4400–5800 K, rather than converging on a single temperature point. This is not a methodological artifact, but rather reflects the fact that **different physical quantities are sensitive to different stages of convection zone development.** The emergence and deepening of the convection zone is a continuous physical process, beginning at approximately 5800 K and completing at approximately 4400 K. Different physical quantities are triggered at different thresholds during this process, constituting five independent time stamps:

Table 9: Five Time Stamps of Convection Zone Development and Their Physical Correspondence to the Five-Dimensional Ruptures

Time Stamp	Temperature	Convection Zone State	Physical Process	Strongest Rupture Dimension	Chow F (4762 K)
1	~5800 K	Extremely thin	One-dimensional LTE approximation begins to fail	Dim. 2 (Model Self-Consistency)	6.29
2	~5500 K	Begins to influence radiation environment	Planetary atmospheric photoevaporation intensifies	Dim. 4 (Radius–Mass Relationship)	27.65
3	~4762 K	Precise K/G spectral type boundary	Molecular opacity dominates, limb darkening qualitative change	Dim. 3 (Measurement Method Divergence)	10.18
4	~4582 K	α - Ω dynamo fully operational	Magnetic braking intensity reaches critical threshold	Dim. 5 (Age–Rotation Relationship)	8.35
5	~4400 K	Completely dominates photosphere	Global limb-darkening properties undergo thorough switch	Dim. 1 (Transit Depth)	648.70

Time Stamp 1: ~5800 K—Stellar model self-consistency begins to rupture. The convection zone is extremely thin, but the one-dimensional LTE approximation already produces detectable deviations at the photospheric boundary. The statistical treatment of the Opacity Distribution Function (ODF) incurs increased error as molecules begin to appear. Dimension 2 ruptures most strongly here.

Time Stamp 2: ~5500 K—The planetary radius–mass relationship begins to rupture. The emergence of the convection zone alters the stellar angular momentum loss rate and the intensity of early ultraviolet radiation. Ultraviolet radiation drives photoevaporation of planetary atmospheres, and the degree of atmospheric retention directly affects the final planetary radius. Dimension 4 ruptures most strongly here.

Time Stamp 3: ~4762 K—Measurement method divergence reaches its maximum. The precise K/G spectral type boundary. Molecular opacity (TiO, VO absorption bands) has become sufficient to dominate the visible and near-infrared bands; the wavelength dependence of limb darkening undergoes its most dramatic switch here. The detector response functions of different facilities capture different

wavelength bands, and the systematic divergence among them is sharply amplified at this location. Dimension 3 ruptures most strongly here.

Time Stamp 4: ~4582 K—The age-rotation relationship rupture is strongest. The convection zone is sufficiently deep, the α - Ω dynamo is fully operational, and the magnetic braking intensity has reached a critical threshold. The rotational evolution switches from "weak braking, slow deceleration" to "strong braking, fast deceleration." Dimension 5 ruptures most strongly here.

Time Stamp 5: ~4400 K—The transit depth equation global rupture is strongest. The convection zone completely dominates the photosphere, molecular opacity holds absolute dominance across all wavelength bands. Stellar atmosphere models face ODF errors throughout the entire photospheric depth range; the global properties of limb darkening undergo their most thorough switch. Dimension 1 ruptures most strongly here.

These five time stamps are not five different "breakpoints"—they are **different signatures left by the same physical process—the emergence and deepening of the convection zone—on different physical quantities**. The Tang Break is not a precise single temperature point, but a temperature zone of approximately 4400–5800 K. 4762 K is the most representative critical point within this zone—it is the precise K/G spectral type boundary, the location where measurement method divergence reaches its maximum, and it passes the structural break test in all five dimensions.

3.9 The Physical Boundary of the Tang Break: The Null Discovery of Vein 4

Vein 4 tested whether structural parameters such as the period ratios of multi-planet systems are modulated by the metallicity of the host star. A total of 769 multi-planet systems were included in the analysis. All three metallicity group structural break tests were non-significant (Poor vs. Solar $F = 0.02$, $p = 0.996$; Poor vs. Rich $F = 1.75$, $p = 0.158$; Solar vs. Rich $F = 1.92$, $p = 0.125$).

This is not a failure, but rather the physical boundary of the Tang Break. It tells us that regime coupling has a clear physical boundary—it acts on physical processes that depend on the stellar radiation field and internal structure, but not on orbital structures that are primarily determined by gravitational dynamics. The gravitational interactions in planetary systems are wavelength-independent, facility-independent, and model-independent—this is one of the few truly "regime-independent" physical quantities.

Table 10: The Physical Boundary of the Tang Break

Modulated by the Tang Break	Not Modulated by the Tang Break
Transit depth (radiation field)	Planetary system orbital architecture (gravitational dynamics)
Stellar model self-consistency (atmospheric physics)	—
Measurement method consistency (radiation field dependence)	—
Planetary radius–mass relationship (planet formation + radiation field)	—
Stellar age–rotation relationship (magnetic braking + evolution)	—

4. Discussion

4.1 The Physical Origin of the Tang Break: Emergence and Deepening of the Convection Zone

Why the K/G transition zone (~4400–5800 K)? Why is 4762 K the most representative critical point within it?

When the effective temperature of a star drops below approximately 5800 K, an outer convection zone begins to emerge. The emergence and deepening of the convection zone triggers a series of systematic physical consequences:

(1) Opacity mechanism switch—onset of molecular dominance (beginning ~5800 K, fully dominant at ~4762 K). In hot stars, opacity is dominated by the H^- negative hydrogen ion, whose wavelength dependence is relatively smooth. When the temperature drops below approximately 5000 K, molecules such as TiO and VO begin to form in large quantities; molecular absorption bands are extremely dense, covering a broad range from the visible to the near-infrared. The wavelength dependence of molecular opacity is far more complex than that of atomic opacity. At approximately 4762 K (the precise K/G spectral type boundary), molecular opacity has become sufficient to dominate the visible and near-infrared bands. One-dimensional LTE models face a fundamental difficulty here: the statistical

treatment of molecular spectral lines (the ODF) has far larger approximation errors at low temperatures than at high temperatures.

(2) Limb-darkening property qualitative change (most dramatic at ~4762 K). The wavelength dependence of limb darkening is entirely determined by the physical mechanism of opacity. In the atomic-opacity-dominated regime, limb darkening varies relatively smoothly with wavelength. In the molecular-opacity-dominated regime, limb darkening varies extremely dramatically with wavelength—some bands are nearly completely opaque, others relatively transparent. When the wavelength dependence of limb darkening undergoes its most dramatic switch at the K/G boundary, the detector response functions of different facilities capture different wavelength bands, and the systematic divergence among them is sharply amplified. **This is the physical root of why Dimension 3 (measurement method divergence) is strongest at approximately 4762 K.**

(3) Magnetic field generation mechanism switch and magnetic braking jump (critical threshold reached at ~4582 K). Once the convection zone is fully developed, convective motions couple with rotation to produce an α - Ω dynamo, generating magnetic fields far stronger than those of radiation-dominated stars. The coupling of the stellar wind with the magnetic field leads to a drastic increase in angular momentum loss (magnetic braking). Kraft (1967) discovered the sharp transition in rotation velocity at approximately 6250 K—this is the magnetic braking effect when the convection zone has "just emerged." Dimension 5 of the Tang Break is strongest at approximately 4582 K—this is a further jump in magnetic braking intensity after the convection zone has become "fully dominant." **Dimension 5 captures the second magnetic braking critical point after the Kraft Break.**

(4) Planetary formation environment switch (strongest at ~5500 K). The emergence of the convection zone alters the early evolutionary trajectory of the star—the angular momentum loss rate, ultraviolet radiation intensity, and stellar wind strength are all different. These factors directly influence the photoevaporation rate of gas in the protoplanetary disk and the degree of retention of planetary atmospheres. When the convection zone is sufficiently deep and magnetic braking is sufficiently strong, the early radiation environment of the star undergoes a qualitative change, and the final planetary radius–mass relationship switches accordingly. **This is the physical root of why Dimension 4 (planetary radius–mass relationship) is strongest at approximately 5500 K.**

(5) Global failure of stellar atmosphere models (most thorough at ~4400 K). When the convection zone completely dominates the photosphere, molecular opacity holds absolute dominance across all wavelength bands. One-dimensional LTE models face ODF errors throughout the entire photospheric depth range. The global properties of limb darkening—not only the wavelength dependence, but also the variation with depth—undergo their most thorough switch here. **This is the ultimate physical root of why Dimension 1 (transit depth) ruptures most strongly at approximately 4400 K.**

The transmission pathway is now complete: Stellar mass determines core temperature → core temperature determines outer convection zone depth → the convection zone emerges and progressively deepens (~4400–5800 K) → opacity switches from atomic-dominated to molecular-dominated (~4762 K) → wavelength dependence of limb darkening undergoes qualitative change → radiation field structure switches → one-dimensional LTE assumptions of stellar atmosphere models fail → systematic divergence among different measurement methods is amplified → the planetary radius–mass relationship is modulated by stellar properties. Simultaneously: convection zone deepening → α - Ω dynamo starts up → magnetic braking intensity jumps (~4582 K) → rotational evolution trajectory switches.

4.2 The Relationship Between the Tang Break and the Kraft Break

The Kraft Break is a classical phenomenon discovered by Robert Kraft in 1967: F-type and hotter stars rotate very rapidly (typical velocities of 100–200 km/s), while G-type and cooler stars rotate drastically more slowly (typical velocities of 1–10 km/s). This phenomenon was later explained as the enhancement of magnetic braking caused by the emergence of the outer convection zone. The Kraft Break has been repeatedly verified by modern instruments—starspot modulation measurements from the Kepler mission (McQuillan et al., 2013, 2014), direct $v \sin i$ measurements from Gaia DR3, and rotational splitting measurements from asteroseismology—all confirm the sharp transition in rotation velocity at 6250 K.

The Tang Break is structurally isomorphic with the Kraft Break in terms of physical origin—both point to the emergence of the stellar outer convection zone. However, the two correspond to different stages of convection zone evolution, and there are fundamental differences in their evidentiary frameworks.

Table 11: Multi-Dimensional Comparison Between the Tang Break and the Kraft Break

Comparison Dimension	Kraft Break (1967)	Tang Break (This Paper)
Discovery Temperature	~6250 K	~4400–5800 K zone (core point 4762 K)
Number of Verified Dimensions	1 (rotation velocity)	5 (radiation field, model self-consistency, measurement methods, planet coupling, rotational evolution)
Methodology	Mean comparison, correlation analysis	Testability Norms + multi-tool cross-validation for structural breaks
Physical Origin	Convection zone "just emerged"	Convection zone "from emergence to full dominance"
Corresponding Convection Zone Stage	Extremely thin convection zone, magnetic braking begins to strengthen	Complete evolution: molecular opacity dominance → limb-darkening qualitative change → magnetic braking jump → full photospheric convection
Theoretical Framework	Independent empirical discovery	Systematic evidence for the Factor Hierarchy Law in astronomy
Cross-Disciplinary Verification	None	Five major financial markets + five astronomical dimensions + first-principles physics calibration

The relationship between the two is complementary—together they depict the complete physical picture of the stellar outer convection zone from its emergence to its full development. The Kraft Break is the first signal of the convection zone's "emergence" (at ~6250 K), while the Tang Break captures five consecutive signals of the convection zone's "deepening" (across the ~4400–5800 K zone). If the Kraft Break were to be systematically re-examined using the Testability Norms in the future, 6250 K could potentially be precisely located as the upper boundary of a broader regime transition zone—from 6250 K (convection zone just emerged) to 4400 K (convection zone fully dominant), constituting a complete regime transition zone in stellar physics.

4.3 Cross-Disciplinary Evidence for the Factor Hierarchy Law

Tang (2026g) established the Factor Hierarchy Law across five major financial markets and constructed the Testability Norms. Tang (2026h) provided the first cross-disciplinary independent empirical case in astronomical transit depth measurements. Tang (2026i) provided the second independent empirical case in astronomical stellar model testing. Tang (2026k) completed the first-principles physics calibration on the Onsager exact solution of the 2D Ising model, confirming a precise information-theoretic duality between the Testability Norms and the Ehrenfest classification of phase transitions, with an analytical mathematical proof confirming a deviation of strictly zero. This paper integrates the evidence from five independent dimensions into the Tang Break—a regime-switching law of stellar physics itself.

Integrating the five major financial markets (Tang, 2026g), five independent astronomical dimensions (Tang, 2026h, 2026i, and this paper), and the first-principles physics calibration (Tang, 2026k), the Factor Hierarchy Law has now received cross-disciplinary evidentiary support from three completely independent disciplines. This paper considers that there is sufficient evidence to confirm its status as a Law. All current independent verifications originate from the same author's series of preprints; subsequent cross-researcher and cross-data-source independent verification is a necessary condition for the Law to gain broad acceptance within the scientific community.

4.4 The Ontological Significance of the Tang Break: The Regime Boundary of Measurement Itself

The existence of the Tang Break implies a deeper epistemological fact: on either side of the K/G transition zone, physical quantities that depend on stellar atmosphere models—such as effective temperature, surface gravity, and planetary radius—are operationally defined quantities under different regimes, rather than universal objective properties. This is not a shortcoming of model refinement, but an ineliminable regime boundary inherent in measurement itself.

This insight elevates the Tang Break from "a law of stellar physics" to "an ontological verification of the Testability Norms in astronomy." It demonstrates that in complex systems, the answer to the question "what is the value of this datum?" itself depends on "under which regime, using which model, and with which facility it was measured." Without answering these preconditions, any statement about data is an incomplete proposition.

The value of the Testability Norms lies in providing an operational identification procedure for such regime boundaries. The Tang Break is the highest-level evidence produced by this procedure in astronomy—it not only verifies the Factor Hierarchy Law across five independent dimensions, but also reveals the regime limits of measurement itself.

4.5 The Complete 5W2H Closure

The integrative analysis of this paper also constitutes a complete implementation of the Question Completeness dimension (5W2H) at the "cross-dimensional integration" level: **What**—Stars possess a regime boundary (the Tang Break) in the K/G transition zone; **Why**—The physical origin is the emergence and deepening of the convection zone; **Who**—The temperature transition zone of 4400–5800 K and the surface gravity critical point of $\log g = 4.64$ are the physical thresholds of the Tang Break; **Where**—On the main sequence, covering the entire core parameter space of F/G/K/M-type stars; **When**—Throughout the main-sequence evolutionary stage of stars, continuously operative during the entire process from the emergence to the full dominance of the stellar outer convection zone; **How**—The transmission pathway runs from core temperature to outer convection, to opacity switching and limb-darkening qualitative change, to magnetic braking jump, to all measurements and physical relationships that depend on the radiation field; **How Much**—The structural break test F-values across the five dimensions range from 6.29 to 648.70, all with $p < 0.001$, all passing permutation or matched-sample tests.

4.6 Citation Chain Transparency Statement and Directions for Subsequent Verification

Citation chain transparency. The core evidence of this paper—Dimension 1 from Tang (2026h), Dimension 2 from Tang (2026i), Dimensions 3 through 5 from the preprints of this series—is all based on the same public raw data file (39,913 rows $\times$ 355 columns) from the NASA Exoplanet Archive, and all were completed by the same author. Although the five dimensions use different dependent variables, different data subsets, and different physical mechanisms, constituting logically independent verification, all current verification originates from the same researcher's series of preprints. Cross-researcher and cross-data-source independent verification remains to be completed. This paper candidly declares this limitation and clearly states that the final establishment of the Tang Break and the Factor Hierarchy Law requires independent verification by other researchers using different datasets and independent code implementations. The complete methodological descriptions and public data sources provided in this paper offer a fully actionable pathway for such cross-researcher verification.

Directions for subsequent verification. (1) **Gaia independent photometric verification:** The internal cross-validation of stellar atmosphere model self-consistency (Dimension 2) requires external

independent verification from Gaia's trigonometric parallaxes and independent photometric data. (2) **High-resolution spectroscopic verification:** The physical explanation of molecular opacity dominance requires direct verification from high-resolution spectroscopy—measuring the strength of molecular absorption bands on either side of the Tang Break to confirm the quantitative relationship with the radiation field switch. (3) **Cluster age–rotation verification:** The sample size for the age–rotation relationship (Dimension 5) is limited (498 stars); stellar rotation distributions in star clusters of different ages could provide stronger large-sample verification for the Tang Break in the evolutionary dimension. (4) **Multi-breakpoint physical mechanism verification:** Targeted physical mechanism verification should be conducted for the strongest rupture points of different physical quantities at different positions within the transition zone, to establish a precise correspondence between convection zone evolutionary stages and the regime switches of each physical quantity. (5) **Cross-disciplinary independent verification:** The final establishment of the Factor Hierarchy Law requires isomorphic independent verification in further disciplines (chemical catalysis, meteorology, seismology, etc.).

5. Conclusion

This paper has integrated the complete chains of evidence from five independent dimensions across Tang (2026h, 2026i) and this series of studies, yielding the following conclusions:

(1) **The existence of the Tang Break:** Stars possess a universal "regime boundary" at temperatures of approximately 4400–5800 K (the K/G transition zone) and a surface gravity of approximately $\log g = 4.64$. This boundary corresponds to the complete physical process of the stellar outer convection zone from its emergence to its full dominance. Five entirely independent physical dimensions all undergo structural ruptures within this transition zone. Cross-validation using multiple structural break testing tools rules out the possibility of any single-tool bias.

(2) **The physical origin of the Tang Break:** The emergence and deepening of the convection zone. The convection zone begins to emerge at approximately 5800 K and completely dominates the photosphere by approximately 4400 K. This continuous process sequentially triggers stellar model deviations, planetary radius–mass relationship switches, measurement method divergence amplification, rotational evolution trajectory switches, and global transit depth equation rupture. 4762 K is the most representative critical point within this transition zone—it is the precise K/G spectral type boundary, the location where molecular

opacity begins to fully dominate limb darkening, passes the structural break test in all five dimensions, and receives the most precise independent verification in Dimension 3 (threshold regression 4723 K, Bai-Perron 4750 K).

(3) **The relationship between the Tang Break and the Kraft Break:** Both share the same physical origin (convection zone emergence), but the Kraft Break captures only a single-dimensional single effect (~6250 K, rotation velocity), whereas the Tang Break captures five-dimensional effects and the complete evolutionary process (~4400–5800 K zone). The two are complementary, together depicting the complete physical picture of the stellar outer convection zone from its emergence to its full development. If the Kraft Break were to be systematically re-examined using the Testability Norms in the future, 6250 K could potentially be precisely located as the upper boundary of a broader regime transition zone.

(4) **The physical boundary and ontological significance of the Tang Break:** Regime coupling acts on physical processes that depend on the radiation field and internal structure, but not on planetary orbital structures that are primarily determined by gravitational dynamics. The null discovery of Vein 4 (all metallicity structural break tests non-significant) provides negative evidence for this boundary. The existence of the Tang Break implies that on either side of the K/G transition zone, physical quantities that depend on stellar atmosphere models are operationally defined quantities under different regimes. This is a physical limit of measurement itself, not a model defect.

(5) **The cross-disciplinary establishment of the Factor Hierarchy Law:** Integrating the five major financial markets (Tang, 2026g), five independent astronomical dimensions (Tang, 2026h, 2026i, and this paper), and the first-principles physics calibration (Tang, 2026k), the Factor Hierarchy Law has now received cross-disciplinary evidentiary support from three completely independent disciplines. This paper considers that there is sufficient evidence to confirm its status as a Law. Subsequent cross-researcher and cross-data-source independent verification is a necessary condition for the Law to gain broad acceptance within the scientific community.

The Tang Break is not a statistical game, but a stellar physical law that uses statistics as its tool and physics as its foundation. It leaves detectable statistical signatures across five independent dimensions during the process of the convection zone from emergence to full dominance. The Testability Norms allow these signatures to emerge automatically from the data; physics endows them with unified meaning. The ultimate value of the Tang Break lies in this: it not only verifies the Factor Hierarchy Law, but also reveals the regime limits of measurement itself—in complex systems, the answer to the question "what is the value of this datum?" itself depends on "under which regime, using which model, and with which facility it was measured."

Data Availability Statement

All data in this series of studies were obtained from the NASA Exoplanet Archive (<https://exoplanetarchive.ipac.caltech.edu>), downloaded via the TAP synchronous query interface. The data are fully public and require no application. Methodological details are provided in the original analyses of each dimension; any researcher can independently reproduce all results.

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